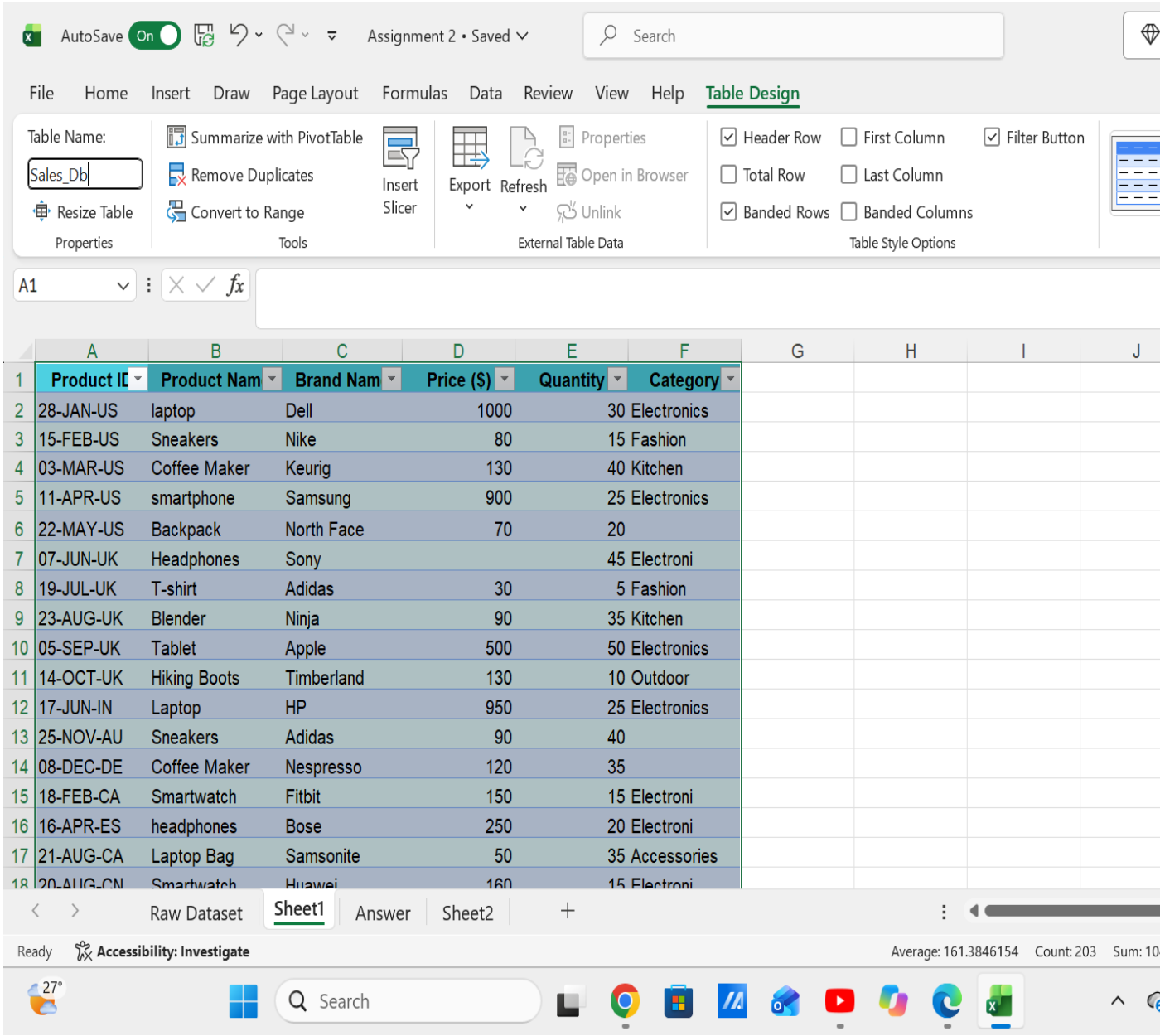
Data Cleaning Using Excel Power Query

STEP – 1 Converting raw dataset into table and given Table Name “Sales _DB”



Product ID	Product Nam	Brand Nam	Price ($)	Quantity	Category
28-JAN-US	laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	smartphone	Samsung	900	25	Electronics
22-MAY-US	Backpack	North Face	70	20	
07-JUN-UK	Headphones	Sony		45	Electroni
19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	
08-DEC-DE	Coffee Maker	Nespresso	120	35	
18-FEB-CA	Smartwatch	Fitbit	150	15	Electroni
16-APR-ES	headphones	Bose	250	20	Electroni
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electroni

STEP – 2 Handling Missing Values in Excel

There are two tasks:

1. Find and handle missing values in the Price ($) column.
2. Find and handle missing values in the Category column.

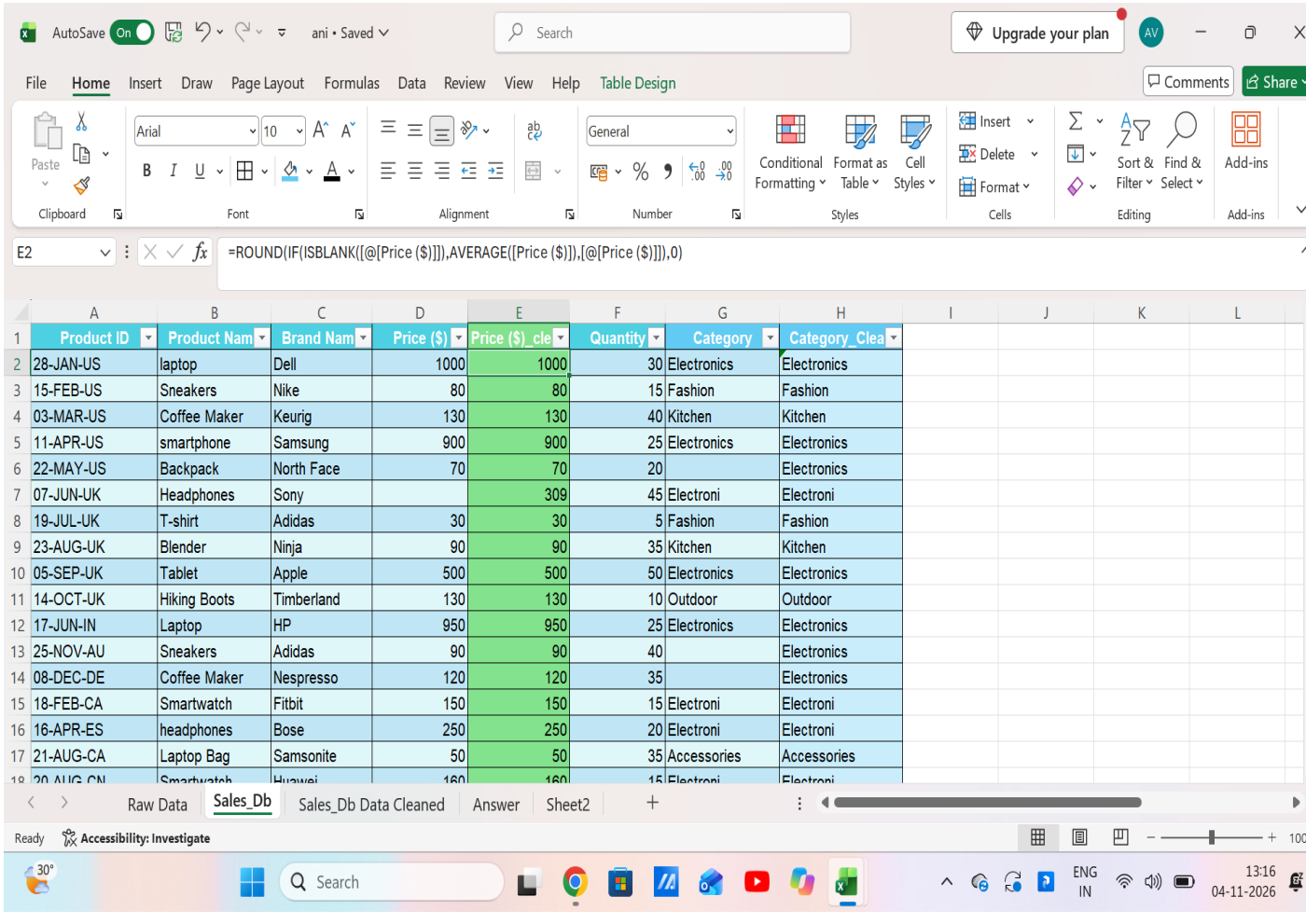
A. Check for missing values in the Price ($) column

Step 1: Find missing values

1. Select the Price ($) column.
2. Click the filter dropdown and select the blank.
3. Click ok. Now blanked cells are showed the column.
4. Create a new column named **Price Cleaned**
5. Apply the AVERAGE formula:

   = (IF (ISBLANK ([@ [Price ($)]]), AVERAGE ([Price ($)]), [@ [Price ($)])

6. Finally missed **Price Cleaned** Column Filled.



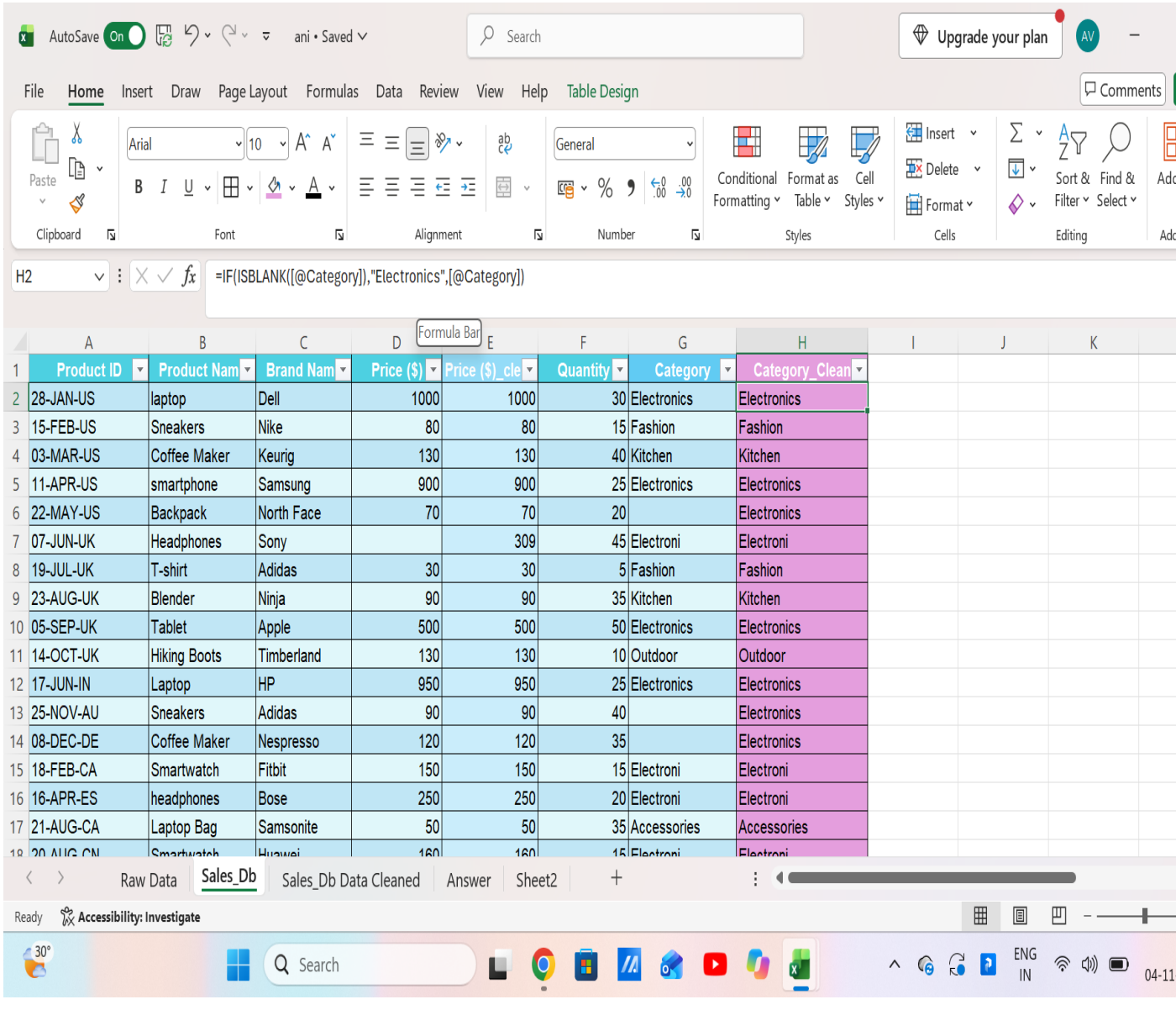
	A	B	C	D	E	F	G	H
1	Product ID	Product Nam	Brand Nam	Price ($)	Price ($)_cle	Quantity	Category	Category_Clea
2	28-JAN-US	laptop	Dell	1000	1000	30	Electronics	Electronics
3	15-FEB-US	Sneakers	Nike	80	80	15	Fashion	Fashion
4	03-MAR-US	Coffee Maker	Keurig	130	130	40	Kitchen	Kitchen
5	11-APR-US	smartphone	Samsung	900	900	25	Electronics	Electronics
6	22-MAY-US	Backpack	North Face	70	70	20		Electronics
7	07-JUN-UK	Headphones	Sony		309	45	Electroni	Electroni
8	19-JUL-UK	T-shirt	Adidas	30	30	5	Fashion	Fashion
9	23-AUG-UK	Blender	Ninja	90	90	35	Kitchen	Kitchen
10	05-SEP-UK	Tablet	Apple	500	500	50	Electronics	Electronics
11	14-OCT-UK	Hiking Boots	Timberland	130	130	10	Outdoor	Outdoor
12	17-JUN-IN	Laptop	HP	950	950	25	Electronics	Electronics
13	25-NOV-AU	Sneakers	Adidas	90	90	40		Electronics
14	08-DEC-DE	Coffee Maker	Nespresso	120	120	35		Electronics
15	18-FEB-CA	Smartwatch	Fitbit	150	150	15	Electroni	Electroni
16	16-APR-ES	headphones	Bose	250	250	20	Electroni	Electroni
17	21-AUG-CA	Laptop Bag	Samsonite	50	50	35	Accessories	Accessories

B. Check for missing values in the Category column

Step 1: Find missing categories

1. Select the Category column.
2. Click the filter dropdown and select the blank.
3. Click ok. Now blanked cells are showed the column.
4. Create a new column named **Category Cleaned**
5. Apply the IF formula:

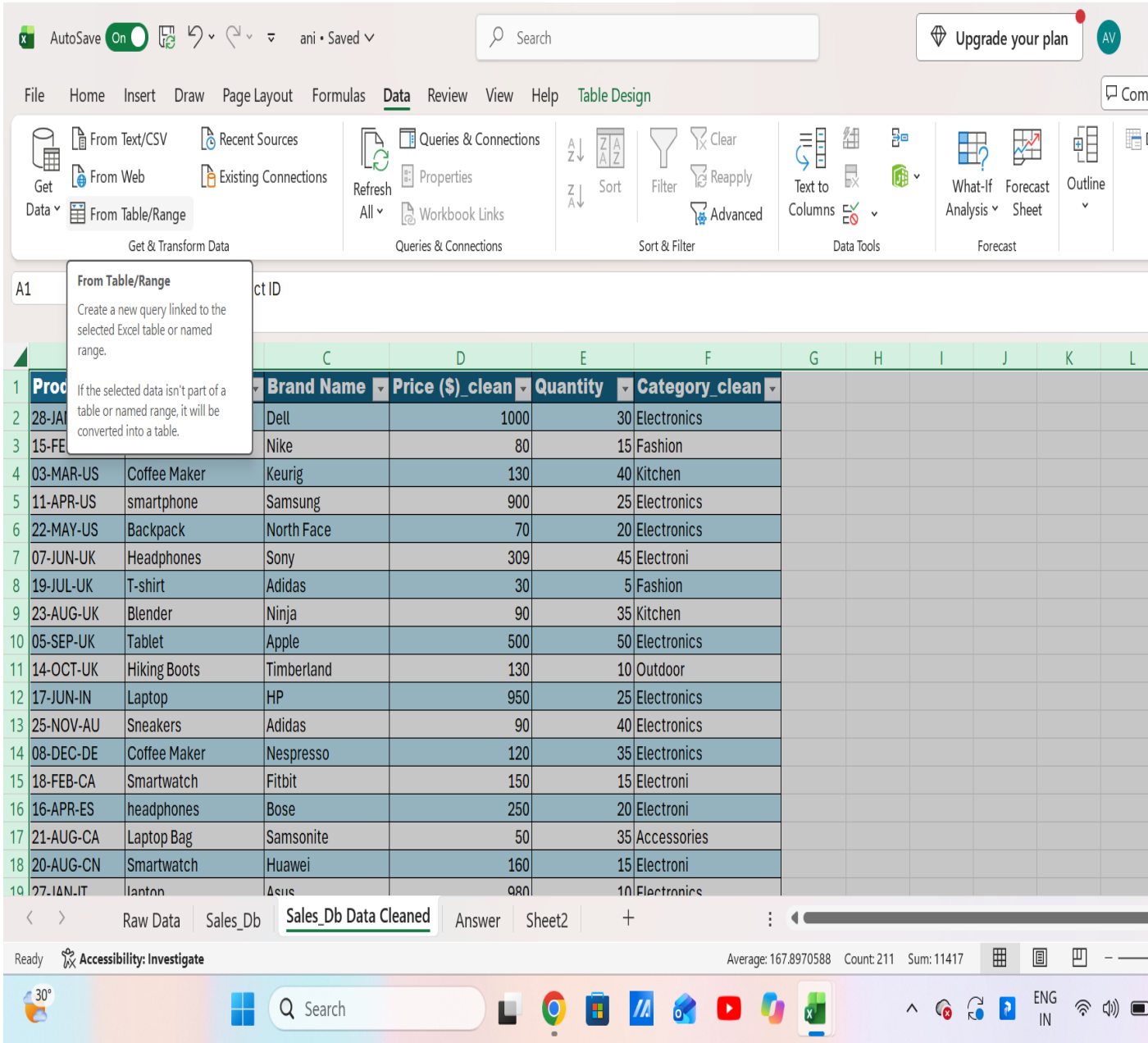
   =IF(ISBLANK([@Category]),"Electronics", [@Category])
6. Finally missed Category Cleaned column Filled.



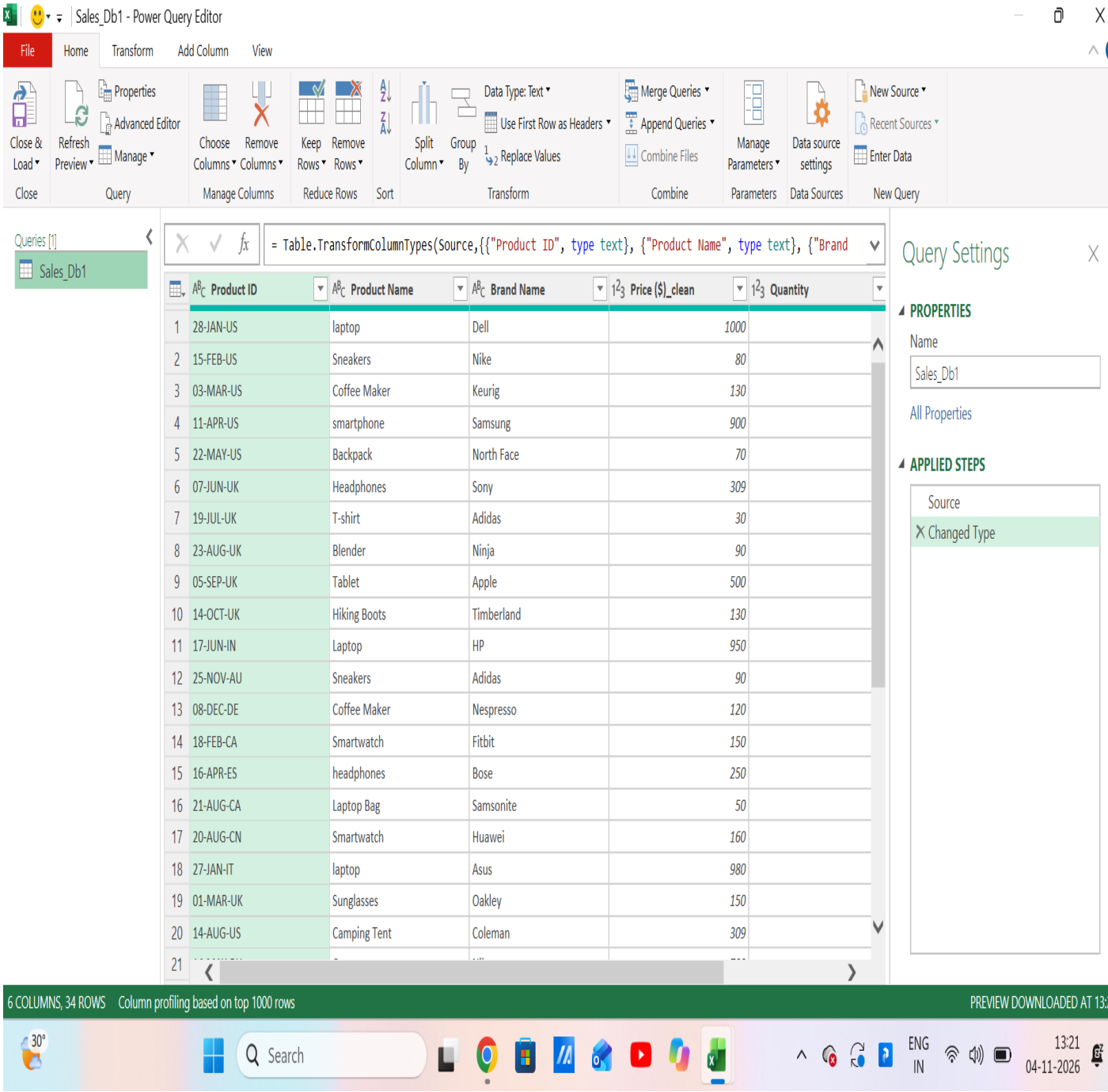
	A	B	C	D	E	F	G	H
1	Product ID	Product Nam	Brand Nam	Price ($)	Price ($)_cle	Quantity	Category	Category_Clean
2	28-JAN-US	laptop	Dell	1000	1000	30	Electronics	Electronics
3	15-FEB-US	Sneakers	Nike	80	80	15	Fashion	Fashion
4	03-MAR-US	Coffee Maker	Keurig	130	130	40	Kitchen	Kitchen
5	11-APR-US	smartphone	Samsung	900	900	25	Electronics	Electronics
6	22-MAY-US	Backpack	North Face	70	70	20		Electronics
7	07-JUN-UK	Headphones	Sony		309	45	Electroni	Electroni
8	19-JUL-UK	T-shirt	Adidas	30	30	5	Fashion	Fashion
9	23-AUG-UK	Blender	Ninja	90	90	35	Kitchen	Kitchen
10	05-SEP-UK	Tablet	Apple	500	500	50	Electronics	Electronics
11	14-OCT-UK	Hiking Boots	Timberland	130	130	10	Outdoor	Outdoor
12	17-JUN-IN	Laptop	HP	950	950	25	Electronics	Electronics
13	25-NOV-AU	Sneakers	Adidas	90	90	40		Electronics
14	08-DEC-DE	Coffee Maker	Nespresso	120	120	35		Electronics
15	18-FEB-CA	Smartwatch	Fitbit	150	150	15	Electroni	Electroni
16	16-APR-ES	headphones	Bose	250	250	20	Electroni	Electroni
17	21-AUG-CA	Laptop Bag	Samsonite	50	50	35	Accessories	Accessories

Step 3: Load Data

1. Select the dataset in Excel.
2. Go to Data → From Table/Range.



Opened the Power Query Editor.



	Product ID	Product Name	Brand Name	Price ($)_clean	Quantity
1	28-JAN-US	laptop	Dell	1000	
2	15-FEB-US	Sneakers	Nike	80	
3	03-MAR-US	Coffee Maker	Keurig	130	
4	11-APR-US	smartphone	Samsung	900	
5	22-MAY-US	Backpack	North Face	70	
6	07-JUN-UK	Headphones	Sony	309	
7	19-JUL-UK	T-shirt	Adidas	30	
8	23-AUG-UK	Blender	Ninja	90	
9	05-SEP-UK	Tablet	Apple	500	
10	14-OCT-UK	Hiking Boots	Timberland	130	
11	17-JUN-IN	Laptop	HP	950	
12	25-NOV-AU	Sneakers	Adidas	90	
13	08-DEC-DE	Coffee Maker	Nespresso	120	
14	18-FEB-CA	Smartwatch	Fitbit	150	
15	16-APR-ES	headphones	Bose	250	
16	21-AUG-CA	Laptop Bag	Samsonite	50	
17	20-AUG-CN	Smartwatch	Huawei	160	
18	27-JAN-IT	laptop	Asus	980	
19	01-MAR-UK	Sunglasses	Oakley	150	
20	14-AUG-US	Camping Tent	Coleman	309	
21					

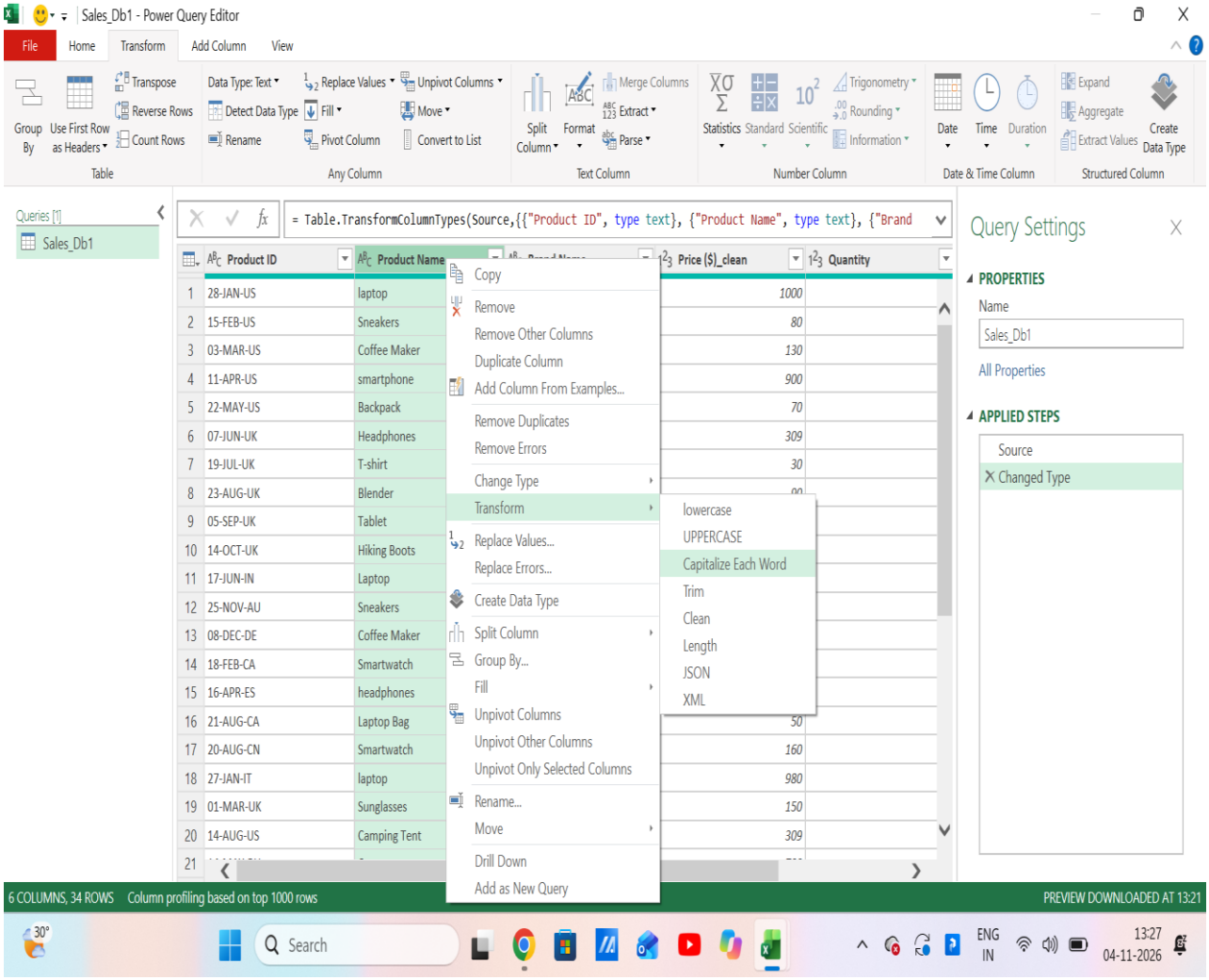
Step 4: Correcting Inconsistent Data

A. Identify inconsistent Product Names

In the dataset, the following inconsistency is present:

How to correct using Power Query:

1. Select the Product Name column.
2. Go to Transform → Format → Capitalize Each Word.
3. This changes the first letter of each word to uppercase.



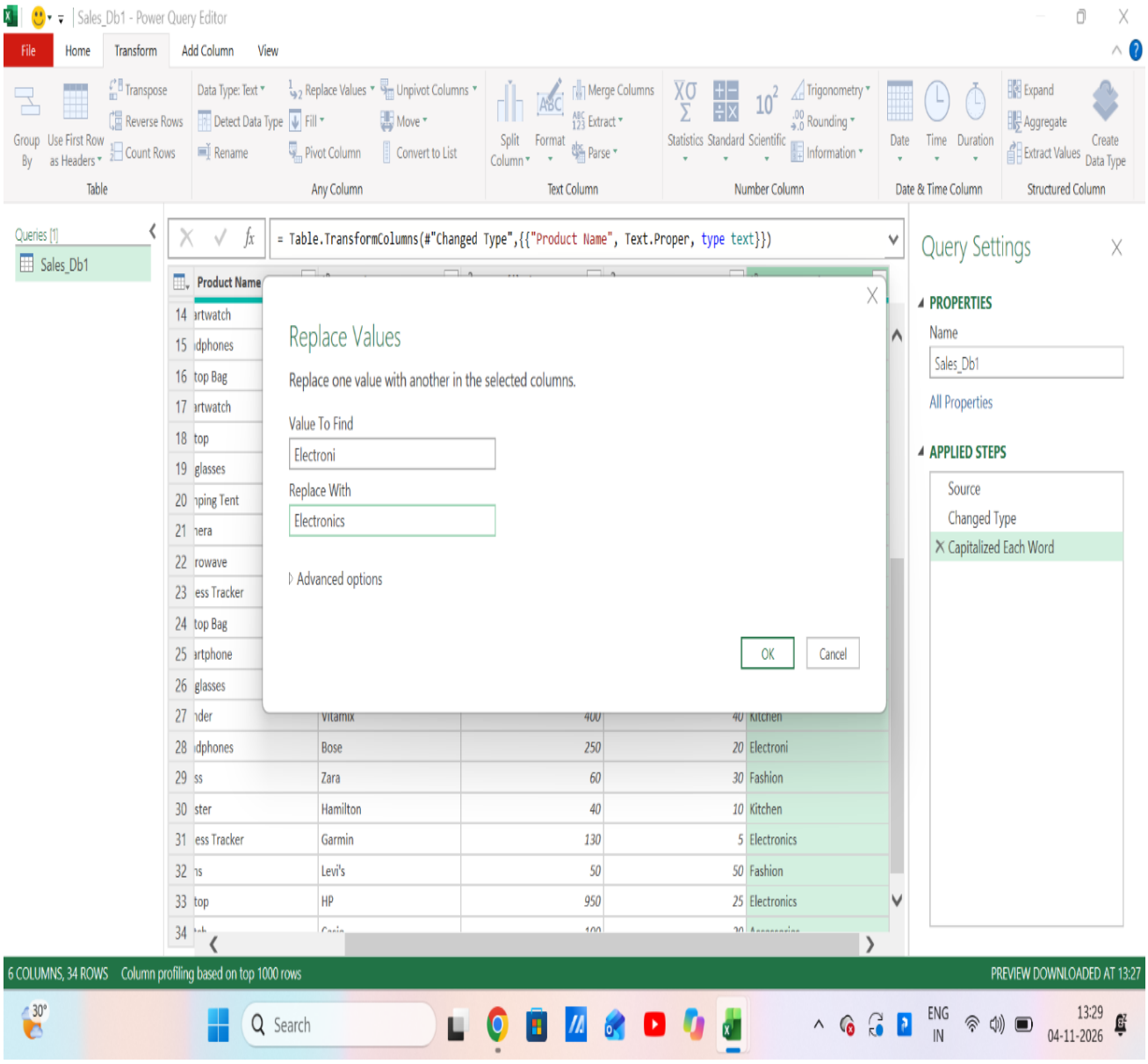
Results:

Original Product Name	Corrected Product Name
headphones	**H**eadphones
Sunglasses	**S**unglasses
Laptop	**L**aptop
Camera	**C**amera

B. Correct Category typos

How to correct using Replace Values:

1. Select the Category column.
2. Go to Transform → Replace Values.
3. Enter Electroni in Value to Find.
4. Enter Electronics in Replace With. Click OK.
5. Typos is Electroni

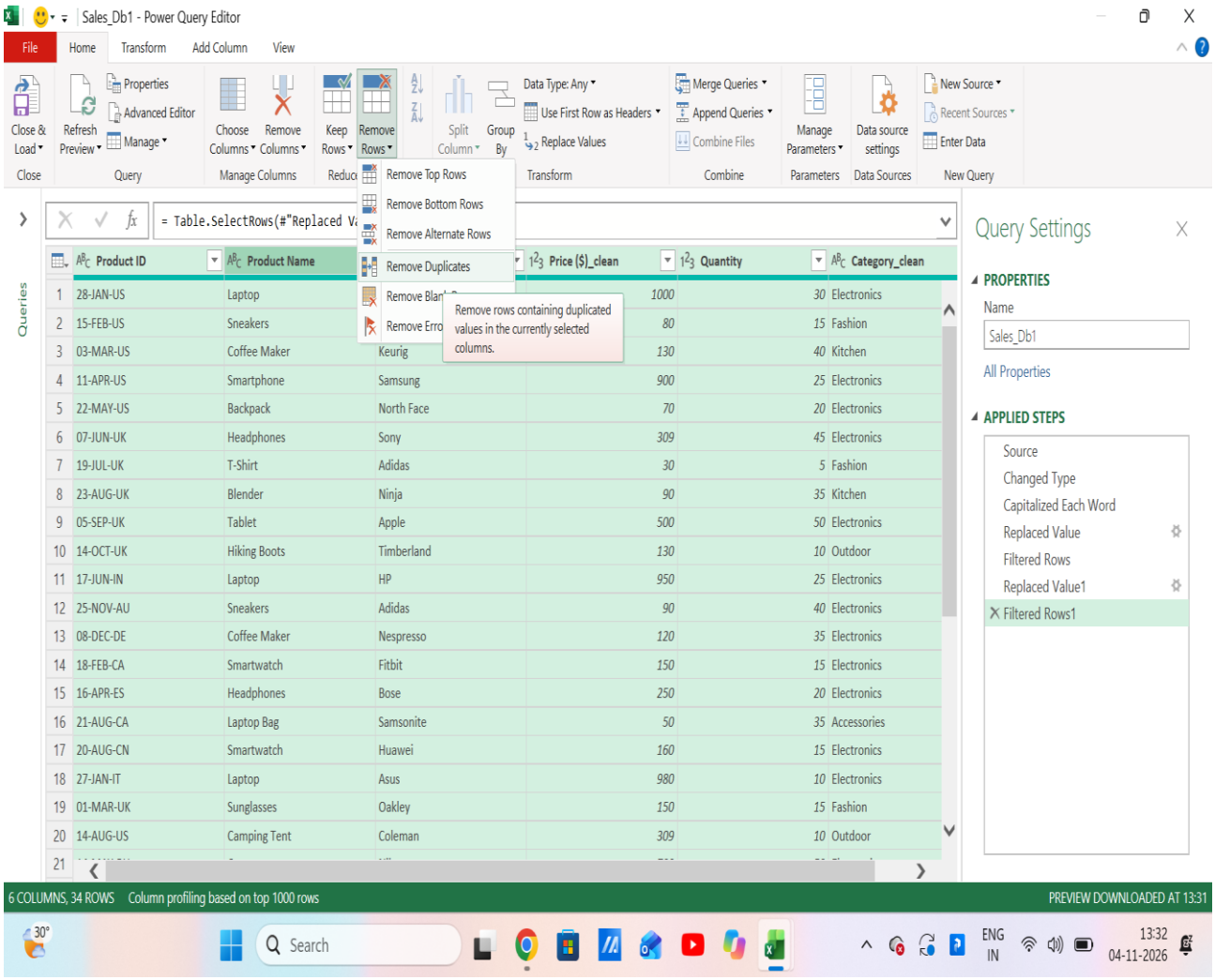


Results: Electroni = Electronics

Step 5: Removing Duplicates

How to remove duplicate rows:

1. Select all six columns using Ctrl + A (inside the table preview, ensure all columns are selected).
2. Go to Home → Remove Rows → Remove Duplicates.
3. Power Query checks all selected columns together.
4. Duplicate rows will be removed.



Results: Total 34 Original Rows -3 duplicate Rows =31 Unique Rows

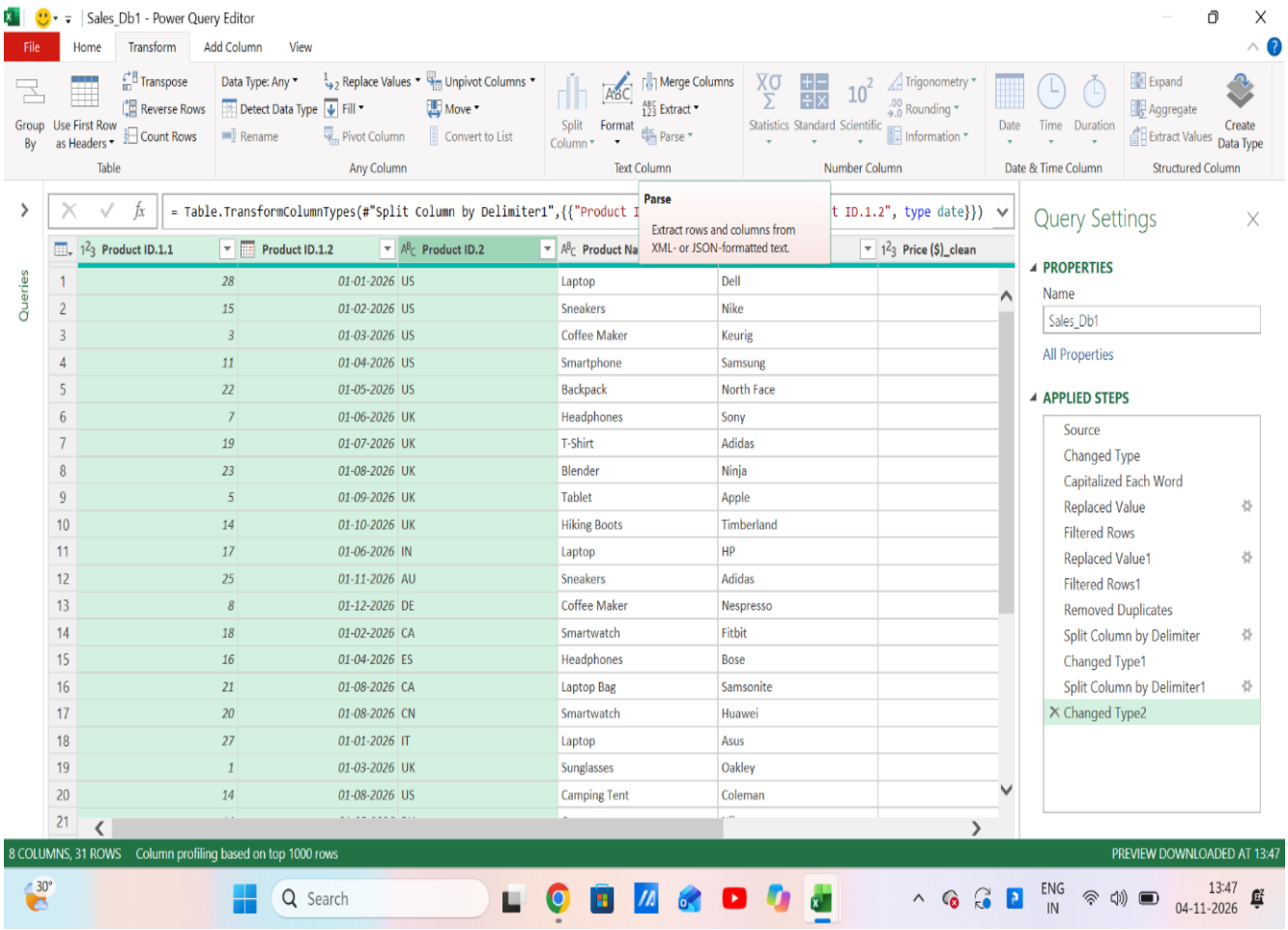
Step 6: Splitting and Merging Data

A. Split Product ID into Manufacturing Date and Country Code

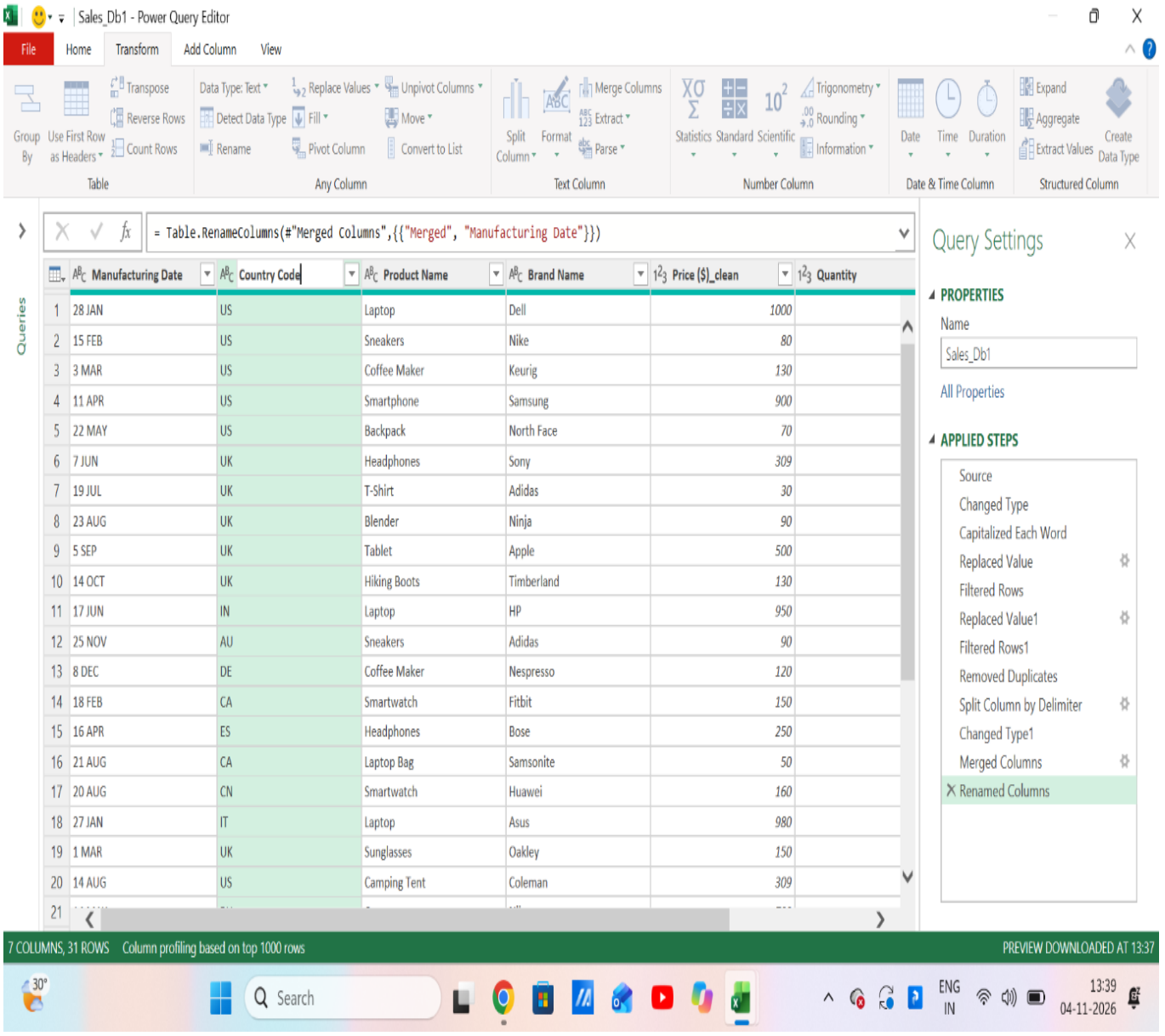
How to split:

1. Select the Product ID column.
2. Go to Transform → Split Column → By Delimiter.
3. Select delimiter: -.
4. Select Each occurrence of the delimiter.
5. Click OK.
6. Three columns will appear.
7. Rename them:
    - Manufacturing Day
    - Manufacturing Month
    - Country Code
8. Select the first two columns and use Merge Columns with - as the separator.
9. Rename the merged column to **Manufacturing Date.**

Here Columns are Splitter into 3 Columns.



Here First 2 Columns are Merged (Manufacturing Date) and 3rd Column name is Renamed (Country Code).

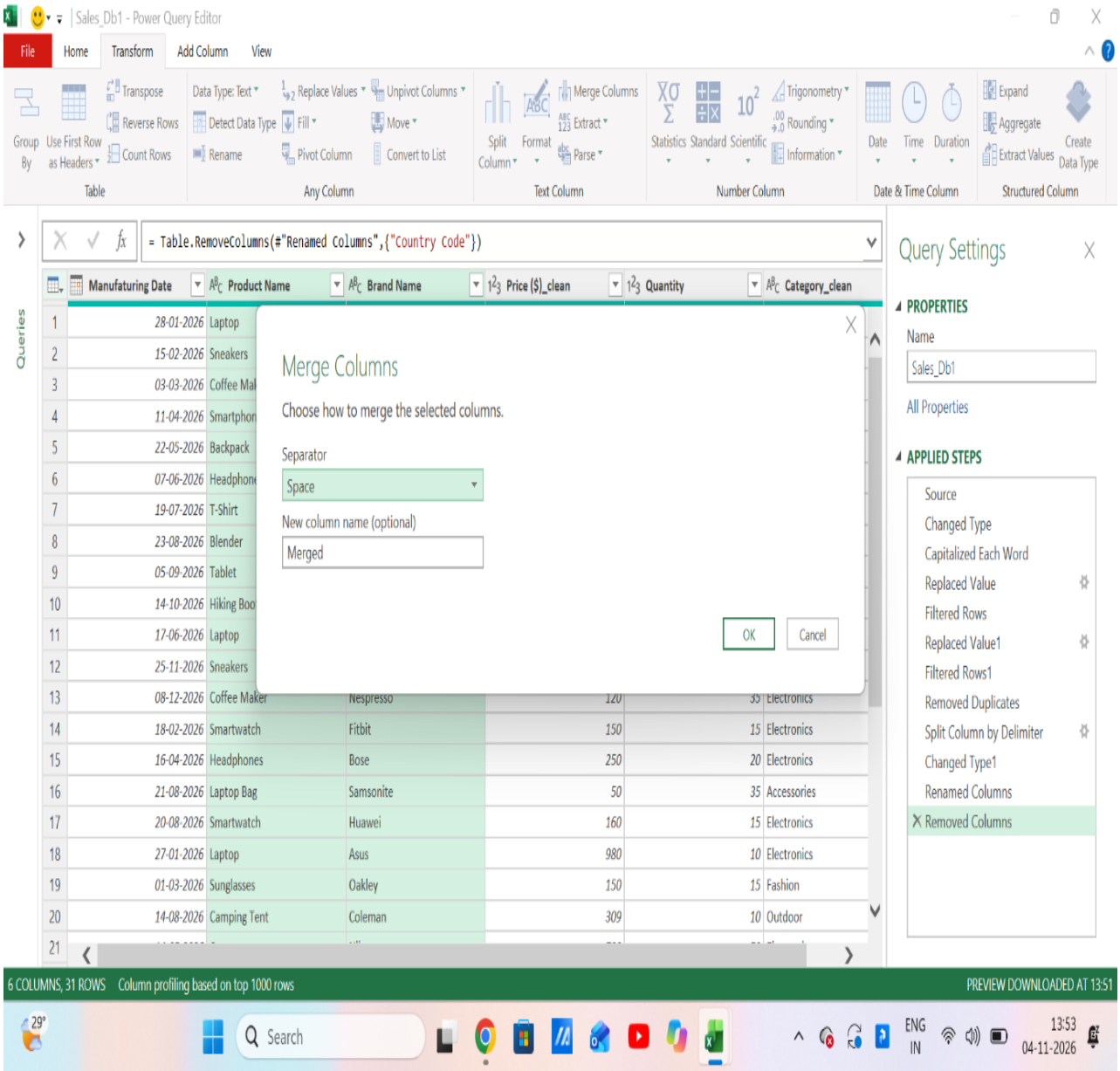


	Manufacturing Date	Country Code	Product Name	Brand Name	Price ($)_clean	Quantity
1	28 JAN	US	Laptop	Dell	1000	
2	15 FEB	US	Sneakers	Nike	80	
3	3 MAR	US	Coffee Maker	Keurig	130	
4	11 APR	US	Smartphone	Samsung	900	
5	22 MAY	US	Backpack	North Face	70	
6	7 JUN	UK	Headphones	Sony	309	
7	19 JUL	UK	T-Shirt	Adidas	30	
8	23 AUG	UK	Blender	Ninja	90	
9	5 SEP	UK	Tablet	Apple	500	
10	14 OCT	UK	Hiking Boots	Timberland	130	
11	17 JUN	IN	Laptop	HP	950	
12	25 NOV	AU	Sneakers	Adidas	90	
13	8 DEC	DE	Coffee Maker	Nespresso	120	
14	18 FEB	CA	Smartwatch	Fitbit	150	
15	16 APR	ES	Headphones	Bose	250	
16	21 AUG	CA	Laptop Bag	Samsonite	50	
17	20 AUG	CN	Smartwatch	Huawei	160	
18	27 JAN	IT	Laptop	Asus	980	
19	1 MAR	UK	Sunglasses	Oakley	150	
20	14 AUG	US	Camping Tent	Coleman	309	
21						

B. Merge Brand Name and Product Name

How to merge:

1. Select Brand Name.
2. Hold Ctrl and select Product Name.
3. Go to Transform → Merge Columns.
4. Select Space as the separator.
5. Enter the new column name: Product Brand.
6. Click OK.



Result: Brand Name + Product Name = Product Brand

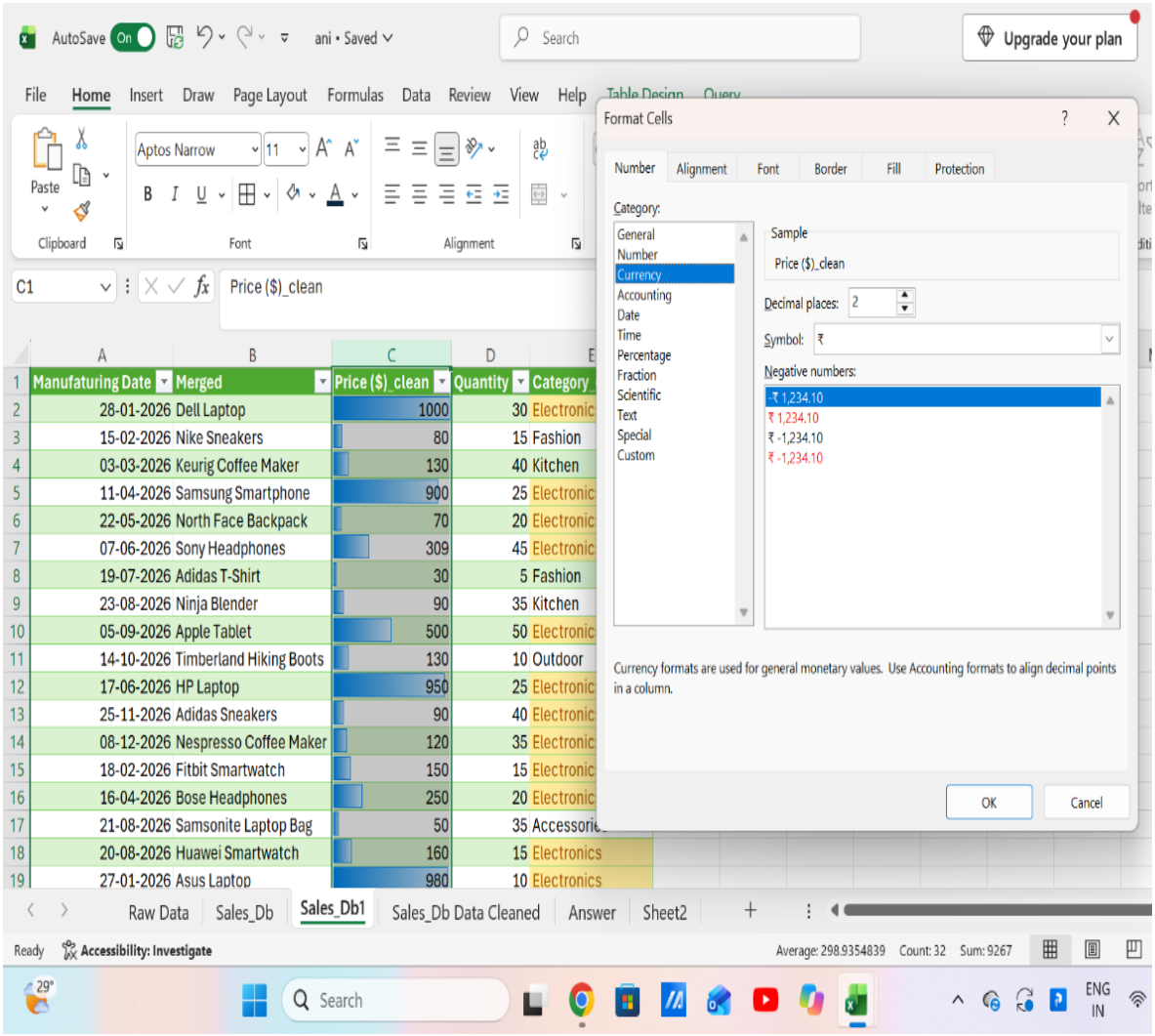
Ex: Dell + Laptop = Dell Laptop

Step 7: Number Formatting

A. Format the Price column to Currency

Steps in Excel:

1. Select the Price ($) column.
2. Go to the Home tab.
3. Find the Number group.
4. Click the Number Format dropdown.
5. Select Currency.
6. Click OK.

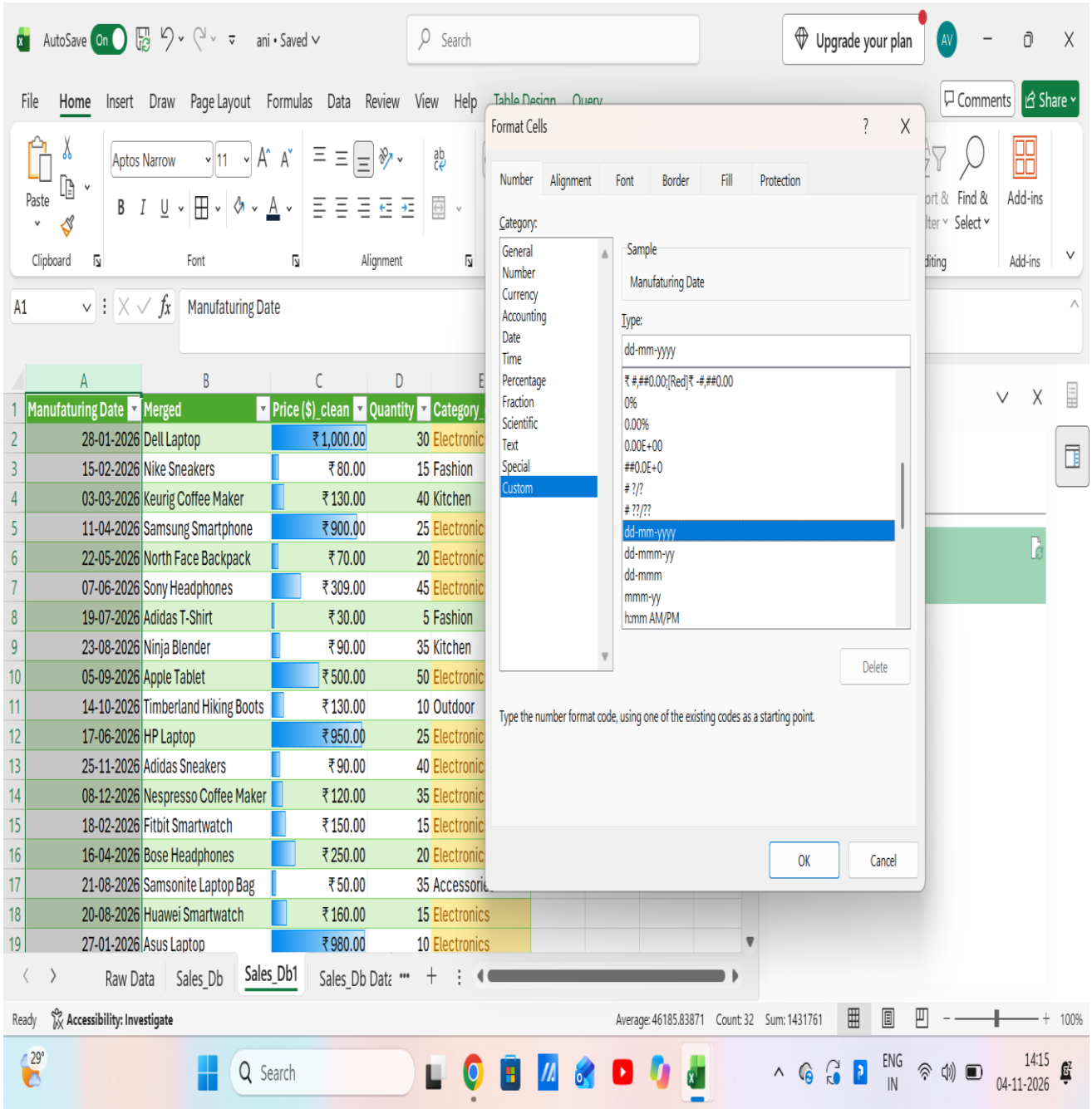


Result: All available prices are displayed in currency format. ($1000)

B. Format Manufacturing Date as DD-MM-YYYY

Steps in Excel:

1. Select the Manufacturing Date column.
2. Right-click and select Format Cells.
3. Select the Number tab.
4. Choose Custom and Select the DD-MM-YYYY and Click ok.

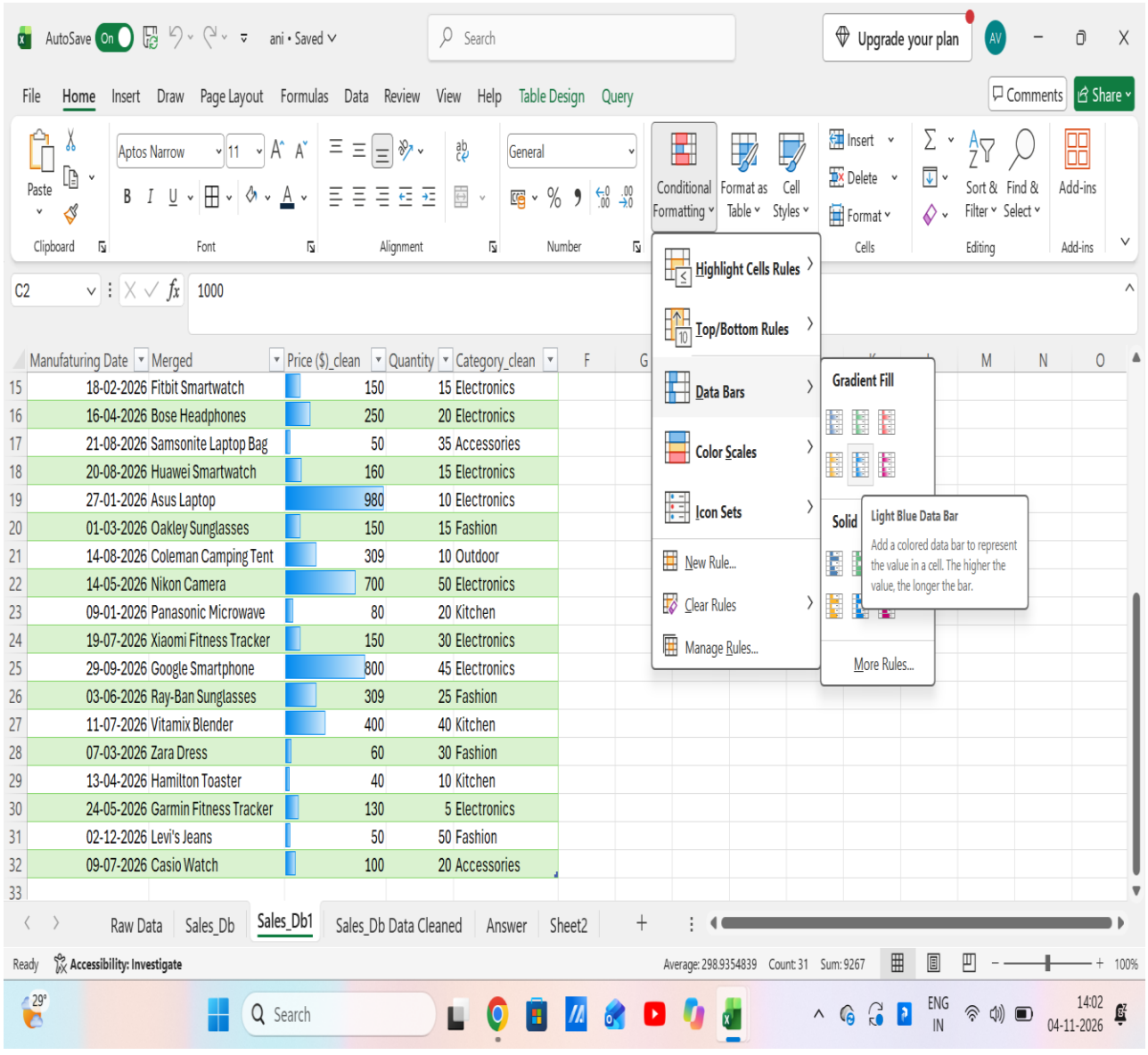


Step 8: Conditional Formatting

1.Apply Data Bars to the Price column

Steps in Excel:

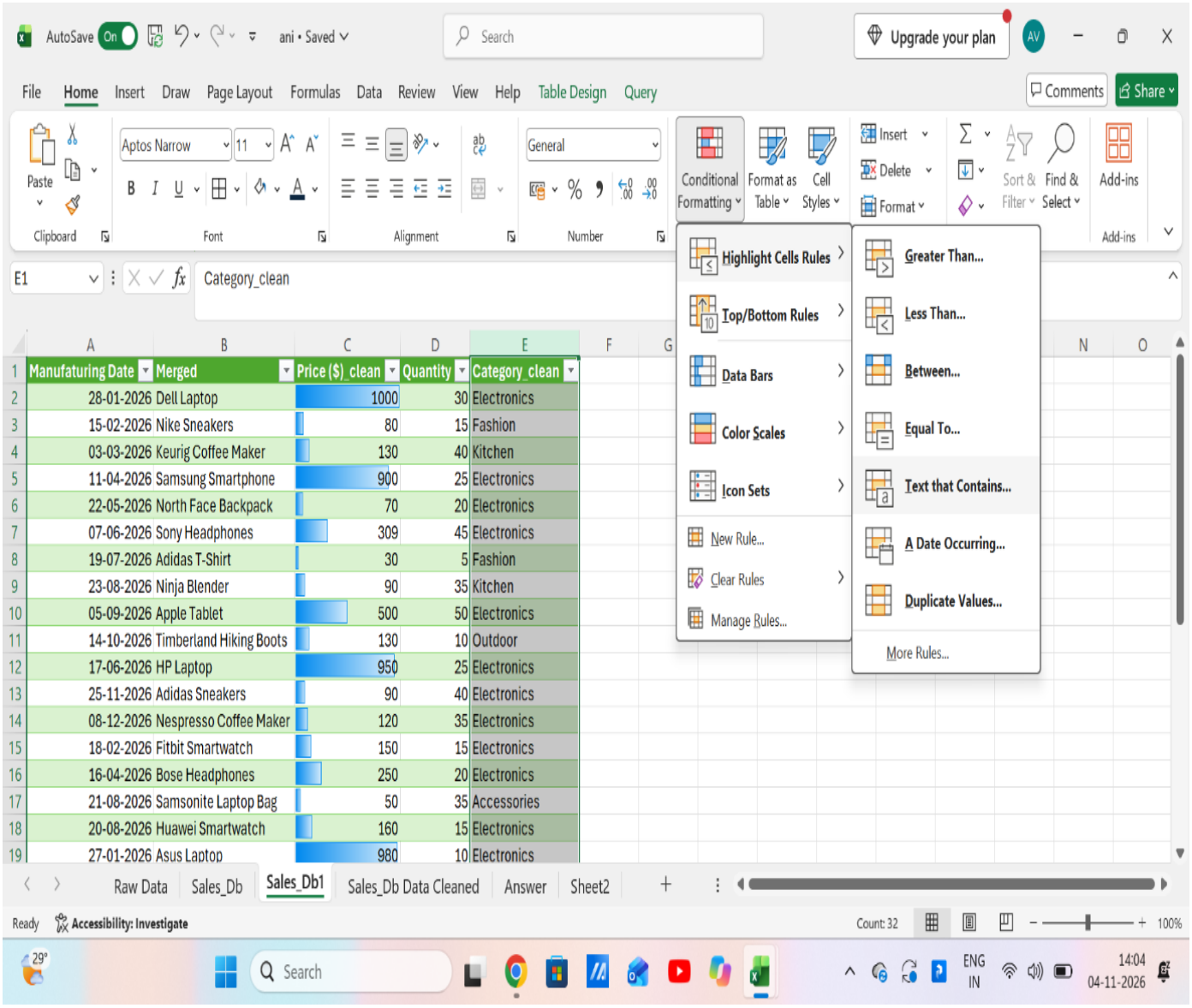
1. Select the Price ($) column.
2. Go to the Home tab.
3. Click Conditional Formatting.
4. Select Data Bars.
5. Choose Solid Fill or Gradient Fill.
6. Select your preferred colour.



B. Create a Custom Conditional Formatting Rule for Electronics

Steps in Excel:

1. Select all cells in the Category column.
2. Go to the Home tab.
3. Click Conditional Formatting.
4. Select Highlight Cells Rules → Text That Contains.
5. Type Electronics.
6. Choose Light Red Fill with Dark Red Text or another colour.
7. Click OK.



Manufaturing Date	Merged	Price ($)_clean	Quantity	Category_clean
28-01-2026	Dell Laptop	1000	30	Electronics
15-02-2026	Nike Sneakers	80	15	Fashion
03-03-2026	Keurig Coffee Maker	130	40	Kitchen
11-04-2026	Samsung Smartphone	900	25	Electronics
22-05-2026	North Face Backpack	70	20	Electronics
07-06-2026	Sony Headphones	309	45	Electronics
19-07-2026	Adidas T-Shirt	30	5	Fashion
23-08-2026	Ninja Blender	90	35	Kitchen
05-09-2026	Apple Tablet	500	50	Electronics
14-10-2026	Timberland Hiking Boots	130	10	Outdoor
17-06-2026	HP Laptop	950	25	Electronics
25-11-2026	Adidas Sneakers	90	40	Electronics
08-12-2026	Nespresso Coffee Maker	120	35	Electronics
18-02-2026	Fitbit Smartwatch	150	15	Electronics
16-04-2026	Bose Headphones	250	20	Electronics
21-08-2026	Samsonite Laptop Bag	50	35	Accessories
20-08-2026	Huawei Smartwatch	160	15	Electronics
27-01-2026	Asus Laptop	980	10	Electronics

Result: Finally, Electronics showed in Yellow Colour